JEE Mathematics: Quick Review with Grading and Practice Map - Quadratic Equation (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (45 questions; a question that appears in two of these is counted once): High = 4 or more questions, Medium = 1 to 3 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Standard Form of Quadratic Equation

Question 01

Can you write down the formula for the roots of a quadratic equation?

Model Answer

For $ax^2+bx+c=0,\ a\neq 0$: $x=\dfrac{-b\pm\sqrt{D}}{2a}$, where $D=b^2-4ac$ is the discriminant.

$$\alpha=\frac{-b+\sqrt{D}}{2a},\quad \beta=\frac{-b-\sqrt{D}}{2a}$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

Explain the condition(s) about the nature of roots.

Model Answer

$D>0$: real and distinct; $D=0$: real and equal; $D<0$: imaginary (a conjugate pair when coefficients are real).

Roots real $\Rightarrow D\geq 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 2, 5; C-I 11, 15; C-II 15 | DPP: 1(1-4, 6); 2(13-15, 24); 4(17); 5(5)

Also used in: Sheet: J-II 7-8, 36; J-M 21; J-A 3; C-I 9, 19; C-II 9 | Question Bank: 7

Relation between Roots and Coefficients

Question 03

Write the result for the sum, product and difference of roots.

Model Answer

$$\alpha+\beta=-\frac{b}{a},\qquad \alpha\beta=\frac{c}{a},\qquad |\alpha-\beta|=\frac{\sqrt{D}}{|a|}$$

Ex: $x^2-5x+3=0$: $\alpha+\beta=5,\ \alpha\beta=3,\ |\alpha-\beta|=\sqrt{13}$.

Grade - High

Practice map, main concept: Sheet: J-II 1, 44; J-A 3; C-II 13, 16 | DPP: 1(7, 13-15, 18, 20); 2(16); 4(7, 15) | Question Bank: 3

Also used in: Sheet: J-I 1, 3, 5, 7-12, 14; J-II 3, 45, 47; J-M 2, 8, 18, 20; C-I 1, 6, 9-10, 12, 14, 20, 22; C-II 1, 11 | DPP: 1(8-12, 16-17, 19, 22-24); 2(17-19, 23, 25-26, 28) | Question Bank: 2, 4, 7-10, 14

Question 04

When do imaginary roots occur in conjugate pairs?

Model Answer

When $a, b, c \in \mathbb{R}$ and $a \neq 0$: if $p + iq$ is a root, then $p - iq$ is the other root.

Ex: $x^2 - 4x + 5 = 0$ has roots $2 \pm i$. This need not hold if the coefficients are not all real.

Grade - High

Practice map, main concept: none

Also used in: Sheet: J-I 19; J-M 7, 12; J-A 2, 8 | DPP: 2(7-9, 20-22); 4(2) | Question Bank: 20-21

Question 05

Explain the condition(s) for rational roots of a quadratic equation.

Model Answer

For $a, b, c \in \mathbb{Q}$: if D is a perfect square of a rational number, both roots are rational.

If D is not a perfect square and $p + \sqrt{q}$ (p rational, $\sqrt{q}$ a surd) is a root, the other root is $p - \sqrt{q}$.

Ex: rational coefficients, one root $2 + \sqrt{3} \Rightarrow$ other root $2 - \sqrt{3}$, equation $x^2 - 4x + 1 = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-II 2

Also used in: Sheet: J-M 17; J-A 4 | DPP: 1(5); 2(4-6, 12); 4(1)

Question 06

When the roots are required to be integers or natural numbers, how do you use the sum and product of roots?

Model Answer

Use $\alpha + \beta$ and $\alpha\beta$ to list the factor pairs of the product, test which pairs give the required sum, and use divisibility conditions on $\alpha\beta$ and $\alpha + \beta$ to rule out the rest.

Ex: $x^2 - 70x + \lambda = 0$ has natural roots α, β and $2, 3 \nmid \lambda$. Then α, β are odd and not multiples of 3, with $\alpha + \beta = 70$: $\alpha = 1$ gives $\beta = 69$ (multiple of 3, rejected), $\alpha = 3$ rejected, $\alpha = 5$ gives $\beta = 65$. So $\lambda_{\min} = 325$.

Grade - Medium

Practice map, main concept: Sheet: J-M 18; C-I 10

Also used in: Sheet: J-I 7, 11

Symmetric Function of Roots

Question 07

Can you identify whether a given expression is symmetric in two or more variables?

Model Answer

An expression is symmetric if it is unchanged when any two of the variables are interchanged.

Symmetric: $\alpha^2 + \beta^2$, $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$, $\dfrac{\alpha}{\beta} + \dfrac{\beta}{\alpha}$. Not symmetric: $\alpha^2 - \beta^2$, $\alpha^2\beta$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 6

Question 08

Are you able to write a symmetric expression in terms of the sum and product?

Model Answer

$$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \qquad \alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$$
$$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \qquad (\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$$
$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}, \qquad \frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$$

Ex: $x^2-5x+3=0$: $\alpha^2+\beta^2=19$, $\alpha^3+\beta^3=80$, $\alpha^4+\beta^4=343$.

Grade - High

Practice map, main concept: Sheet: J-I 3, 6; J-II 3, 7, 47; J-M 8; J-A 6; C-I 9, 12, 22; C-II 1 | DPP: 1(17, 19, 21-24); 2(17-19, 25) | Question Bank: 2, 4, 9

Also used in: Sheet: J-I 8-10, 14; J-II 1; J-M 22; C-I 6 | DPP: 2(23, 27) | Question Bank: 3

Question 09

Are you able to solve questions on a quadratic equation when a relation between the roots is given?

Model Answer

Write the given relation (e.g. $\alpha=2\beta$, $\alpha-\beta=3$, $\alpha=\beta^2$) together with $\alpha+\beta=-\frac{b}{a}$ and $\alpha\beta=\frac{c}{a}$, and eliminate α,β.

Ex: $x^2-7x+k=0$, $\alpha=2\beta$: $3\beta=7\Rightarrow\beta=\frac{7}{3}$, $k=2\beta^2=\dfrac{98}{9}$.

Ex: $x^2-11x+k=0$, $\alpha=\beta+3$: $2\beta+3=11\Rightarrow\beta=4$, $k=4\cdot7=28$.

Grade - Medium

Practice map, main concept: Sheet: J-I 7, 9, 12; C-I 1 | DPP: 1(8-12, 16); 2(26)

Also used in: Sheet: J-II 16; J-M 11; C-II 13 | DPP: 1(13-15, 18); 2(16) | Question Bank: 19

Question 10

How do you find $\alpha^n+\beta^n$ when the roots are complex conjugates (use of polar form)?

Model Answer

If $D<0$, write $\alpha=re^{i\theta}$, $\beta=re^{-i\theta}$ with $r=\sqrt{\alpha\beta}$ and $\cos\theta=\dfrac{\alpha+\beta}{2r}$. Then $\alpha^n+\beta^n=2r^n\cos n\theta$.

Ex: $x^2-3x+3=0$: $r=\sqrt{3}$, $\cos\theta=\frac{\sqrt{3}}{2}\Rightarrow\theta=\frac{\pi}{6}$, so $\alpha^{12}+\beta^{12}=2\cdot3^6\cos2\pi=1458$.

Grade - Medium

Practice map, main concept: Sheet: J-M 12 | Question Bank: 20

Also used in: none

Formation of Quadratic Equation

Question 11

When roots are given, can you form the quadratic or cubic equation?

Model Answer

Quadratic: $x^2-(\alpha+\beta)x+\alpha\beta=0$. Cubic: $x^3-S_1x^2+S_2x-S_3=0$ (see the item on S_k).

Ex: roots $2,5$: $x^2-7x+10=0$. Real coefficients and one root $2+i$: $x^2-4x+5=0$. Roots $1,2,3$: $x^3-6x^2+11x-6=0$.

Grade - Low

Practice map, main concept: DPP: 1(5); 2(1-12)

Also used in: Sheet: J-II 2 | DPP: 4(7, 15)

Question 12

When a quadratic equation is given, how do you form another quadratic equation whose roots are symmetric expressions of the roots of the given equation?

Model Answer

Find the sum S and product P of the new roots using symmetric functions, then write $x^2 - Sx + P = 0$.

Ex: $x^2 - 5x + 3 = 0$, new roots $\alpha + 1, \beta + 1$: $S = 7,\ P = \alpha\beta + \alpha + \beta + 1 = 9$, so $x^2 - 7x + 9 = 0$.

New roots α^2, β^2: $S = 19,\ P = 9 \Rightarrow x^2 - 19x + 9 = 0$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 8, 10; J-M 20, 22; C-II 11 | DPP: 2(23, 27-28); 4(5)

Also used in: Sheet: J-I 12-13; J-M 2; C-II 1 | DPP: 4(9-11) | Question Bank: 14

Question 13

Expressing $x^n + \dfrac{1}{x^n}$ in terms of $x + \dfrac{1}{x}$, $n = 2, 3, 4$.

Model Answer

With $t = x + \dfrac{1}{x}$: $x^2 + \dfrac{1}{x^2} = t^2 - 2, \quad x^3 + \dfrac{1}{x^3} = t^3 - 3t, \quad x^4 + \dfrac{1}{x^4} = (t^2 - 2)^2 - 2$

Ex: $x^2 + \dfrac{1}{x^2} = 14,\ x > 0$: $t^2 = 16 \Rightarrow t = 4$, then $x^3 + \dfrac{1}{x^3} = 64 - 12 = 52$.

Grade - **Medium**

Practice map, main concept: Sheet: C-II 4 | DPP: 5(1); 6(1)

Also used in: Sheet: J-M 1; C-I 5 | Question Bank: 13

Question 14

Factorization of a quadratic or cubic polynomial when all roots are given.

Model Answer

$ax^2 + bx + c = a(x - \alpha)(x - \beta), \quad ax^3 + bx^2 + cx + d = a(x - \alpha)(x - \beta)(x - \gamma)$

Putting a value $x = k$ gives $(k - \alpha)(k - \beta)$ directly.

Ex: $x^2 - 4x + 2$: $(1 - \alpha)(1 - \beta) = 1 - 4 + 2 = -1$. $2x^3 - 5x + 9$: $(1 - \alpha)(1 - \beta)(1 - \gamma) = \dfrac{2 - 5 + 9}{2} = 3$.

$x^{100} - 2x + 9$: $\prod(1 - x_i) = 1 - 2 + 9 = 8$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 13, 29; C-I 14 | Question Bank: 8

Also used in: Sheet: J-II 6, 15

Question 15

Condition for a quadratic polynomial to be a perfect square.

Model Answer

$ax^2 + bx + c$ is the square of a real linear expression $\iff$ $a > 0$ and $D = 0$. (If $a < 0$ and $D = 0$ it is the negative of a square.)

Ex: $x^2 - 6x + k$ is a perfect square $\Rightarrow 36 - 4k = 0 \Rightarrow k = 9$.

Grade - **Low**

Practice map, main concept: Sheet: J-II 43; C-I 20

Also used in: none

Question 16

What do you do when the coefficients of a quadratic equation are in cyclic form, such as $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$?

Model Answer

Check the sum of the coefficients: $a(b-c)+b(c-a)+c(a-b) = 0$, so $x = 1$ is a root and the other root is $\dfrac{c(a-b)}{a(b-c)}$.

Equal roots $\Rightarrow$ other root is also $1 \Rightarrow \dfrac{c(a-b)}{a(b-c)} = 1 \Rightarrow 2ac = b(a+c)$, i.e. a, b, c are in H.P.

Similarly $(b-c)x^2 + (c-a)x + (a-b) = 0$ with equal roots gives $2b = a + c$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 17

What do you do when the sum of coefficients of an equation is zero?

Model Answer

If $a+b+c = 0$ then $x = 1$ is a root and the other root is $\dfrac{c}{a}$. If $a-b+c = 0$ then $x = -1$ is a root.

Ex: $(b-c)x^2 + (c-a)x + (a-b) = 0$ has roots 1 and $\dfrac{a-b}{b-c}$.

Grade - Low

Practice map, main concept: Sheet: J-II 45 | DPP: 4(1)

Also used in: none

Question 18

How do you simplify a long polynomial expression in a root α of a given quadratic equation?

Model Answer

Use the equation satisfied by α to replace α^2 (and higher powers) by lower powers.

Ex: $\alpha^2 - 4\alpha + 5 = 0$: $\alpha^2 - 4\alpha + 7 = -5 + 7 = 2$.

Ex: $\alpha^2 = 4\alpha - 5$ gives $\alpha^3 = 4\alpha^2 - 5\alpha = 11\alpha - 20$.

Grade - Medium

Practice map, main concept: Sheet: J-I 1 | DPP: 4(11) | Question Bank: 10

Also used in: none

Question 19

Solving questions involving $S_n = p\alpha^n + q\beta^n$ (Newton's formula).

Model Answer

If α, β are roots of $ax^2 + bx + c = 0$, then $aS_{n+2} + bS_{n+1} + cS_n = 0$ for $S_n = p\alpha^n + q\beta^n$ (any constants p, q).

Ex: $x^2 - x - 1 = 0$: $S_{n+2} = S_{n+1} + S_n$, so $S_{12} = S_{11} + S_{10}$.

Ex: $x^2 - 7x - 1 = 0$, $S_n = \alpha^n + \beta^n$: $S_{21} = 50S_{19} + 7S_{18}$ and $S_{17} = S_{19} - 7S_{18}$, so $\dfrac{S_{21} + S_{17}}{S_{19}} = 51$.

Grade - High

Practice map, main concept: Sheet: J-I 4; J-M 14-15, 21, 23-24; J-A 4-5; C-I 2 | Question Bank: 6

Also used in: DPP: 5(1)

Question 20
How do you find the value of a polynomial at a root of a quadratic equation by long division?

Model Answer
Divide $f(x)$ by the quadratic $q(x)$: $f(x) = q(x)Q(x) + R(x)$. At a root of q, $f = R$.
Ex: $x^2 - 6x + 4 = 0$: $x^4 - 12x^3 + 44x^2 - 48x + 22 = (x^2 - 6x + 4)^2 + 6$, so its value is 6.

Grade - Low
Practice map, main concept: DPP: 2(20-22)
Also used in: Sheet: J-II 2

Transformation of Roots

Question 21
When do you apply the concept of transformation of roots?

Model Answer
When an equation is needed whose roots are $f(\alpha), f(\beta)$ (e.g. $2\alpha,\ \alpha+1,\ \frac{1}{\alpha},\ \alpha^2,\ \alpha^3$), especially if the sum and product of the new roots are hard to find, or the equation is of higher degree.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 22
How do you apply the concept of transformation of roots?

Model Answer
Put $y = f(x)$, solve for $x = g(y)$, and substitute into the given equation: $a\,g(y)^2 + b\,g(y) + c = 0$ is the required equation in y.
Ex: $x^2 - 4x + 2 = 0$. Roots $2\alpha, 2\beta$: $x = \frac{y}{2} \Rightarrow y^2 - 8y + 8 = 0$. Roots $\frac{1}{\alpha}, \frac{1}{\beta}$: $x = \frac{1}{y} \Rightarrow 2y^2 - 4y + 1 = 0$.
Roots α^2, β^2: $x^2 + 2 = 4x \Rightarrow (y+2)^2 = 16y \Rightarrow y^2 - 12y + 4 = 0$.

Grade - Low
Practice map, main concept: Sheet: J-II 15; C-I 6 | DPP: 4(4, 9-10)
Also used in: none

Special Cases on Roots & Identity

Question 23
What is the condition when exactly one root is zero?

Model Answer
$c = 0$ and $b \neq 0$ (with $a \neq 0$). The roots are 0 and $-\dfrac{b}{a}$.
Ex: $(p-1)(p-2)(p-3)x^2 + (p-2)(p-3)x + (p-3)(p-4) = 0$ has exactly one zero root for $p = 4$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 24
What is the condition when both roots are zero?

Model Answer
$b = 0$ and $c = 0$ (with $a \neq 0$): the equation is $ax^2 = 0$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 25
What is the condition when exactly one root is infinite, what does it mean for a straight line, and how does it generalise to an equation of degree n?

Model Answer
$a = 0$ and $b \neq 0$: the equation reduces to $bx + c = 0$ (e.g. $x + 3 = 0$ is a quadratic with one root ∞).

Straight line: if an equation is quadratic in the slope m and its m^2 coefficient vanishes, one slope is infinite, i.e. a vertical line satisfies the condition.

Degree n: exactly k roots are infinite iff the first k coefficients are 0 and the $(k+1)$-th is non-zero.

Ex: for the equation in item 23, exactly one root infinite $\iff p = 1$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 26
What is the condition when both roots are infinite, what does it mean for a straight line, and how does it generalise to an equation of degree n?

Model Answer
$a = 0,\ b = 0,\ c \neq 0$: the equation reduces to $c = 0$ (e.g. $3 = 0$), so both roots are infinite.

Straight line: if both the m^2 and m coefficients vanish, both slopes are infinite (no finite slope works, only vertical lines).

Degree n: all n roots are infinite iff $a_0 = a_1 = \cdots = a_{n-1} = 0$ and $a_n \neq 0$.

Ex: in item 23's equation, $p = 2$ gives $a = 0,\ b = 0,\ c = 2 \neq 0$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 27
What is the condition for an equation to be an identity, and its application?

Model Answer
$ax^2 + bx + c = 0$ is an identity (true for all x) $\iff a = b = c = 0$. A quadratic with more than two distinct roots must be an identity.

Ex: $(p-1)(p-2)(p-3)x^2 + (p-2)(p-3)x + (p-3)(p-4) = 0$ is an identity for $p = 3$.

Ex: $\dfrac{(x-a)(x-b)}{(c-a)(c-b)} + \dfrac{(x-b)(x-c)}{(a-b)(a-c)} + \dfrac{(x-c)(x-a)}{(b-c)(b-a)} = 1$ holds at $x = a, b, c$ (three values), so it is an identity.

Grade - **Low**
Practice map, main concept: Sheet: J-I 15-17
Also used in: none

Writing General Equation

Question 28

When α and β satisfy the same relation, how do you write the general equation having them as roots, and what happens if three distinct values satisfy it?

Model Answer

If $\alpha \neq \beta$ both satisfy $f(t) = 0$ (a quadratic in t), then α, β are exactly its two roots: use $\alpha + \beta = -\frac{b}{a},\ \alpha\beta = \frac{c}{a}$.

Ex: $2\alpha^2 - 5\alpha + 3 = 0,\ 2\beta^2 - 5\beta + 3 = 0,\ \alpha \neq \beta$: $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{5/2}{3/2} = \dfrac{5}{3}$.

If three distinct values satisfy a quadratic, it is an identity. Ex: $a\theta^2 + b\theta + c = 3\theta^2 + 5$ for three distinct θ gives $(a-3)\theta^2 + b\theta + (c-5) = 0$, so $a = 3,\ b = 0,\ c = 5$.

Grade - Low

Practice map, main concept: Sheet: J-I 14

Also used in: DPP: 4(5)

Rational Inequality and Intersection of Intervals

Question 29

Are you able to solve rational inequalities?

Model Answer

Make the RHS zero, make leading coefficients positive, factorise into linear factors, mark zeroes on the number line, put + on the rightmost part and move left, changing sign only across odd-power factors (not across even-power factors). Exclude zeroes of the denominator.

Ex: $(x^2 - x - 6)(x^2 + 6x) > 0 \Rightarrow x(x-3)(x+2)(x+6) > 0 \Rightarrow x \in (-\infty, -6) \cup (-2, 0) \cup (3, \infty)$.

Grade - Medium

Practice map, main concept: DPP: 3(1-40); 6(10)

Also used in: Sheet: J-II 22; J-A 3; C-I 8 | DPP: 6(9)

Question 30

Are you able to solve double inequalities?

Model Answer

Split $a < b < c$ into $a < b$ and $b < c$, solve each, and take the intersection of the solution sets.

Ex: $1 < x^2 - 13x + 37 < 7$: $x^2 - 13x + 36 > 0 \Rightarrow x < 4$ or $x > 9$; $x^2 - 13x + 30 < 0 \Rightarrow 3 < x < 10$. Hence $x \in (3, 4) \cup (9, 10)$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 13; C-II 3

Question 31

How do you solve equations and inequalities containing the modulus of a quadratic expression?

Model Answer

Equations: split at the zeroes of the expression inside $|\ |$, solve in each interval, and keep only roots lying in that interval.

Ex: $|x^2-8x+15|-2x+7=0$. For $x\le 3$ or $x\ge 5$: $x^2-10x+22=0\Rightarrow x=5\pm\sqrt{3}$, only $5+\sqrt{3}$ is valid. For $3<x<5$: $x^2-6x+8=0\Rightarrow x=2,4$, only 4 is valid. Sum of roots $=9+\sqrt{3}$.

Inequalities: $|q(x)|<k \iff -k<q(x)<k$ (a double inequality), and $|q(x)|>k \iff q(x)<-k$ or $q(x)>k$.

Ex: $n\in\mathbb{N}$ with $|n^2-10n+19|<6$: $n^2-10n+25>0$ and $n^2-10n+13<0$, i.e. $n\neq 5$ and $5-2\sqrt{3}<n<5+2\sqrt{3}$. So $n\in\{2,3,4,6,7,8\}$: 6 values.

Grade - Medium

Practice map, main concept: Sheet: J-M 10, 13 | Question Bank: 18

Also used in: Sheet: C-I 5 | DPP: 6(9)

Question 32

How do you solve a question in which the solution set of one inequality must be a subset of the solution set of another?

Model Answer

Write both solution sets as intervals. Inclusion $[r_1,r_2]\subseteq[s_1,s_2]$ means $r_1\ge s_1$ and $r_2\le s_2$. If a parameter changes the order of the endpoints, treat each case separately and check the empty set separately.

Ex: every solution of $x^2-6ax+5a^2\le 0$ is a solution of $x^2-14x+40\le 0$. The second set is $[4,10]$. For $a>0$ the first set is $[a,5a]$, which needs $a\ge 4$ and $5a\le 10$: impossible. For $a\le 0$ the first set is not inside $[4,10]$. So no value of a works.

Grade - Low

Practice map, main concept: Sheet: J-II 22 | DPP: 6(9)

Also used in: none

Condition of Common Root(s)

Question 33

Condition if both roots are common and its interpretation.

Model Answer

$\dfrac{a_1}{a_2}=\dfrac{b_1}{b_2}=\dfrac{c_1}{c_2}$: the two equations are proportional (the same equation).

Ex: $x^2-5x-3=0$ and $2x^2+px+q=0$ have both roots common: $\dfrac{1}{2}=\dfrac{-5}{p}=\dfrac{-3}{q}\Rightarrow p=-10,\ q=-6$.

Ex: real p,q and $x^2-5x+7=0$ has imaginary roots, so a common root forces both common: $p=-10,\ q=14$.

Grade - Low

Practice map, main concept: Sheet: J-I 19 | DPP: 4(13)

Also used in: DPP: 4(4)

Question 34
Condition if one root is common and its interpretation.

Model Answer

$$(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$$

Interpretation: the two polynomials have a common linear factor (their resultant is zero); equivalently, both are factors of one cubic.

Ex: $x^2 - 3x + 2 = 0$ and $x^2 + kx + 8 = 0$ share a root: common root $1 \Rightarrow k = -9$; common root $2 \Rightarrow k = -6$.

Grade - High

Practice map, main concept: Sheet: J-I 11, 18; J-II 4-6, 38; J-M 2, 9; J-A 1; C-II 19 | DPP: 4(12, 14) | Question Bank: 5, 7, 14

Also used in: DPP: 4(8)

Question 35
Use of the elimination method in questions where one root is common.

Model Answer

Let α be the common root. Subtract the equations to eliminate α^2 and get α as a linear function of the parameter, then substitute back (or use cross-multiplication on $\alpha^2, \alpha, 1$).

Ex: $x^2 + bx + c = 0$ and $x^2 + cx + b = 0$ $(b \neq c)$: subtracting gives $(b - c)x = b - c \Rightarrow x = 1$, so $1 + b + c = 0$.

Ex: $P_1 = x^2 + (k-29)x - k$, $P_2 = 2x^2 + (2k-43)x + k$ with a common root: $2P_1 - P_2 = -15x - 3k = 0 \Rightarrow x = -\frac{k}{5}$; putting in P_1: $k(k - 30) = 0$, so $k = 0$ or 30.

Grade - High

Practice map, main concept: DPP: 4(8)

Also used in: Sheet: J-I 18; J-II 4-5, 38; J-M 2, 9; J-A 1; C-II 19 | DPP: 4(12, 14) | Question Bank: 5, 14

Polynomial Function

Question 36
Identify whether a given expression is a polynomial.

Model Answer

A polynomial has only non-negative integer powers of x.

Yes: $4x^7 + 5x^2 + 3$, $x^2 + 3x + 4$. No: $x^2 + \dfrac{1}{x}$, $x + \sqrt{x}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 37
Identify the degree, leading coefficient and absolute term of a polynomial.

Model Answer

For $f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_n$ $(a_0 \neq 0)$: degree n, leading coefficient a_0, absolute (constant) term $a_n = f(0)$. Monic means $a_0 = 1$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 38
Find the zeroes of a polynomial. What do they represent on its graph?

Model Answer
Zeroes are the values of x with $f(x) = 0$; they are the x-intercepts $(x, 0)$ of the graph.
Ex: $f(x) = x^2 - 6x + 8 = (x-2)(x-4)$: zeroes $2, 4$, points $(2, 0), (4, 0)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 39
Are you able to construct the graph of any quadratic polynomial?

Model Answer
Steps: (1) sign of a (opens up/down) and D; (2) zeroes, if real; (3) vertex $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$; (4) y-intercept $(0, c)$; draw a parabola symmetric about $x = -\frac{b}{2a}$.
Ex: $y = x^2 - 8x + 12$: $a = 1,\ D = 16$, zeroes $2, 6$, vertex $(4, -4)$, y-intercept 12.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 40
Are you able to read the graph of a quadratic polynomial: value of y, vertex, sign of D, a, b and c, etc.?

Model Answer
$f(x_0)$ is the y-coordinate of the point of the parabola at $x = x_0$ (above the axis $\Rightarrow f(x_0) > 0$).
$a > 0$ if it opens up. $c = f(0)$ is the y-intercept. $D > 0, = 0, < 0$ according as the graph cuts, touches, or misses the x-axis. Axis $x = -\frac{b}{2a}$, so the sign of b follows from the sign of a and the side of the vertex.
Always positive iff $a > 0,\ D < 0$ (the graph lies entirely above the x-axis).

Grade - Low
Practice map, main concept: Sheet: J-I 20-22; J-II 31-32, 39-40 | DPP: 4(3)
Also used in: none

Question 41
Symmetry of a parabola about its vertex.

Model Answer
$f\left(-\frac{b}{2a} + t\right) = f\left(-\frac{b}{2a} - t\right)$. So if $f(p) = f(q)$ with $p \neq q$, then $p + q = -\frac{b}{a}$, i.e. the axis is $x = \frac{p+q}{2}$.
Ex: $y = x^2 - 2x - 3$: $f(0) = f(2) = -3 \Rightarrow$ axis $x = 1$.

Grade - Low
Practice map, main concept: Sheet: J-II 36
Also used in: Sheet: J-I 20; J-II 37 | DPP: 4(3)

Question 42

Condition for a quadratic polynomial to be always positive or always negative, and its application.

Model Answer

$$ax^2+bx+c>0\ \forall x \iff a>0,\ D<0; \qquad ax^2+bx+c<0\ \forall x \iff a<0,\ D<0$$
$$\geq 0\ \forall x \iff a>0,\ D\leq 0; \qquad \leq 0\ \forall x \iff a<0,\ D\leq 0$$

Ex: $x^2+kx+4>0\ \forall x \Rightarrow k^2-16<0 \Rightarrow -4<k<4$.

Ex: $\dfrac{ax^2+2(a+1)x+9a+4}{x^2+8x+32}<0\ \forall x$: denominator is always positive ($D<0$), so the numerator must be always negative, which needs $a<0$; hence no positive integer a works.

Grade - Medium

Practice map, main concept: Sheet: J-I 25-26; J-II 8-9, 33; J-M 16; C-I 19, 24; C-II 3, 9 | Question Bank: 11

Also used in: Sheet: J-II 10-11, 43; J-A 7; C-I 4

Remainder Theorem

Question 43

How do you use the remainder theorem in questions?

Model Answer

Dividing $f(x)$ by $(x-a)$ leaves remainder $f(a)$; $(x-a)$ is a factor iff $f(a)=0$. Dividing by a quadratic $(x-a)(x-b)$ leaves $R=ux+v$; find u,v from $f(a)$ and $f(b)$.

Ex: $x^{100}-7x^2+10$ divided by $(x-1)$: remainder $f(1)=4$. Divided by x^2-x: $f(0)=10,\ f(1)=4 \Rightarrow R=-6x+10$.

Ex: $f(1)=5,\ f(4)=11$, divided by $(x-1)(x-4)$: $R=2x+3$.

Grade - Low

Practice map, main concept: Sheet: J-II 26

Also used in: Sheet: J-II 2

Range of a Quadratic Polynomial

Question 44

Write the formula for the range of a quadratic polynomial.

Model Answer

$y=ax^2+bx+c$: $a>0 \Rightarrow y\in\left[-\dfrac{D}{4a},\infty\right)$; $a<0 \Rightarrow y\in\left(-\infty,-\dfrac{D}{4a}\right]$

The extreme value occurs at $x=-\dfrac{b}{2a}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 6, 22; J-II 27, 43-44; J-M 21 | Question Bank: 3

Question 45
How do you find the range of a quadratic polynomial by completing the square?

Model Answer
Write $y = a\left(x + \dfrac{b}{2a}\right)^2 - \dfrac{D}{4a}$ and use that the square is ≥ 0.
Ex: $y = x^2 - 2x + 3 = (x-1)^2 + 2 \Rightarrow y \in [2, \infty)$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 27
Also used in: none

Question 46
How do you find the range of a quadratic polynomial by forming a quadratic in x and applying $D \geq 0$?

Model Answer
Put $y = f(x)$, write $ax^2 + bx + (c - y) = 0$. For real x we need $D \geq 0$, which is an inequality in y.
Ex: $y = x^2 - 2x + 3$: $4 - 4(3 - y) \geq 0 \Rightarrow y \geq 2$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 47
How do you find the range of a quadratic polynomial if a condition on x is given?

Model Answer
Find the vertex. If $x = -\frac{b}{2a}$ lies in the given interval, the extreme value is at the vertex and the other bound is at the farther end; otherwise the range lies between the values at the two ends.
Ex: $y = x^2 - 2x - 3$ (vertex $(1, -4)$): $x \in \mathbb{R} \Rightarrow [-4, \infty)$; $x \in [0, 3] \Rightarrow [-4, 0]$; $x \in [-2, 0] \Rightarrow [-3, 5]$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 37; C-I 25; C-II 5
Also used in: Sheet: J-II 7, 27

Range of a Rational Function

Question 48
How do you find the range of a Linear/Linear function?

Model Answer
$y = \dfrac{ax + b}{cx + d}$ $(ad \neq bc)$: solve $x = \dfrac{b - dy}{cy - a}$; x exists unless $cy = a$. So $y \in \mathbb{R} - \left\{\dfrac{a}{c}\right\}$.
Ex: $y = \dfrac{2x + 1}{3x + 5} \Rightarrow y \in \mathbb{R} - \left\{\dfrac{2}{3}\right\}$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 49

How do you find the range of a Linear/Linear function if a condition on x is given?

Model Answer

Solve for x in terms of y, then impose the given condition on x; this gives an inequality in y.

Ex: $y = \dfrac{x-1}{2x+5}$, $x \geq 0$: $x = \dfrac{5y+1}{1-2y} \geq 0 \Rightarrow y \in \left[-\dfrac{1}{5}, \dfrac{1}{2}\right)$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 48

Question 50

How do you find the range of a rational function if there is a common factor?

Model Answer

Cancel the common factor, find the range of the reduced function, then remove the value of y that corresponds to the cancelled x.

Ex: $y = \dfrac{x^2-3x+2}{2x^2+x-3} = \dfrac{(x-1)(x-2)}{(x-1)(2x+3)} = \dfrac{x-2}{2x+3}$, $x \neq 1$: range of the reduced function $\mathbb{R} - \{\frac{1}{2}\}$; at $x = 1$ it gives $-\frac{1}{5}$, so $y \in \mathbb{R} - \left\{\dfrac{1}{2}, -\dfrac{1}{5}\right\}$.

Grade - Low

Practice map, main concept: Sheet: J-II 41

Also used in: none

Question 51

How do you find the range of $\dfrac{Q}{Q}$, $\dfrac{Q}{L}$, $\dfrac{L}{Q}$, $\dfrac{1}{Q}$ if there is no common factor?

Model Answer

Put $y = f(x)$ and cross-multiply to form a quadratic in x. Case 1: leading coefficient $\neq 0$, impose $D \geq 0$ and solve for y. Case 2: put the leading coefficient $= 0$ and check whether that y is attained. For $\dfrac{Q}{L}$ the x^2 coefficient is a constant, so no Case 2 is needed. For $\dfrac{1}{Q}$, take the reciprocal of the range of Q (with $y \neq 0$).

Ex: $y = \dfrac{x^2+x+1}{x^2-x+1}$: $(y-1)x^2 - (y+1)x + (y-1) = 0$, $D \geq 0 \Rightarrow y \in \left[\dfrac{1}{3}, 3\right]$; $y = 1$ gives $x = 0$, so it is included.

Grade - Medium

Practice map, main concept: Sheet: J-I 23-24; J-II 10, 34, 48; J-M 5, 19; C-I 18, 26 | DPP: 5(9-12)

Also used in: Sheet: J-II 33, 35, 41; C-II 3

General Equation of 2nd Degree in Two Variables

Question 52

Memorize the general form of a quadratic expression in two variables.

Model Answer

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 53

What is the condition for the general form of a quadratic expression in two variables to be expressed as a product of 2 linear factors?

Model Answer

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

For real factors, additionally $h^2 \geq ab$ (parallel lines when $h^2 = ab$).

Ex: $x^2 + 3xy + 2y^2 + 2x + 3y + 1$: $a = 1, h = \frac{3}{2}, b = 2, g = 1, f = \frac{3}{2}, c = 1$ gives $\Delta = 0$.

Grade - Low

Practice map, main concept: Sheet: J-I 27 | DPP: 5(4, 13-14)

Also used in: none

Question 54

Both methods to factorize the general form of a quadratic expression in two variables into 2 linear factors.

Model Answer

Method 1: treat it as a quadratic in x; the discriminant is a perfect square in y, and the roots $x = \ldots$ give the factors.

Method 2: factorise the homogeneous part $ax^2+2hxy+by^2$ as $(l_1x+m_1y)(l_2x+m_2y)$, add constants $(l_1x + m_1y + c_1)(l_2x + m_2y + c_2)$, and compare coefficients.

Ex: $x^2 + 3xy + 2y^2 + 2x + 3y + 1 = (x + y + 1)(x + 2y + 1)$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 27 | DPP: 5(4, 13)

Question 55

How do you find the range of x and y if a general equation of 2nd degree in x and y is given?

Model Answer

For the range of y, write it as a quadratic in x and impose $D_x \geq 0$. For the range of x, write it as a quadratic in y and impose $D_y \geq 0$.

Ex: $x^2 + 4y^2 - 3x + 2 = 0$: as a quadratic in x, $D_x = 9 - 4(4y^2 + 2) \geq 0 \Rightarrow y \in \left[-\frac{1}{4}, \frac{1}{4}\right]$; as a quadratic in y, $D_y = -16(x^2 - 3x + 2) \geq 0 \Rightarrow x \in [1, 2]$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 12

Question 56

What do you mean by "for every real value of x there is a real value of y and for every real value of y there is a real value of x"?

Model Answer

It means the range of both x and y is $\mathbb{R}$: $D_y \geq 0$ for every real x and $D_x \geq 0$ for every real y.

Ex: $x^2+3xy+2y^2+2x+3y+1 = 0$: as a quadratic in y, $D_y = (3x+3)^2 - 8(x+1)^2 = (x+1)^2 \geq 0$ for all x; as a quadratic in x, $D_x = (3y + 2)^2 - 4(2y^2 + 3y + 1) = y^2 \geq 0$ for all y.

Grade - Low

Practice map, main concept: Sheet: J-II 12

Also used in: none

Question 57

What is the condition for $ax^2 + 2hxy + by^2$ to be positive for all $(x, y) \neq (0, 0)$, and how is it used?

Model Answer

Treat it as a quadratic in $\frac{x}{y}$ (and in $\frac{y}{x}$): it is positive for all $(x, y) \neq (0, 0) \iff a > 0$ and $h^2 < ab$ (so $b > 0$ too).

Use: the matrix $\begin{pmatrix} a & h \\ h & b \end{pmatrix}$ then has $ab - h^2 > 0$, so the system $ax + hy = p,\ hx + by = q$ has a unique solution.

Ex: $x^2 + 2kxy + 4y^2 > 0$ for all $(x, y) \neq (0, 0) \iff k^2 < 4 \iff -2 < k < 2$.

Grade - Medium

Practice map, main concept: Sheet: J-II 11; J-A 7

Also used in: none

Theory of Equation Involving Higher Degree

Question 58

What do $S_1, S_2, \ldots, S_n$ mean and what is their relation with sum and product?

Model Answer

S_k is the sum of products of the roots taken k at a time. S_1 is the sum of the roots and S_n is their product.

Ex: roots α, β, γ: $S_1 = \alpha + \beta + \gamma,\ S_2 = \alpha\beta + \beta\gamma + \gamma\alpha,\ S_3 = \alpha\beta\gamma$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 28; J-II 13; J-M 11; C-I 13, 16; C-II 14 | DPP: 5(2, 8) | Question Bank: 19

Question 59

How do you find S_i for any n-degree equation?

Model Answer

For $a_0x^n + a_1x^{n-1} + \cdots + a_n = 0$: $S_i = (-1)^i \dfrac{\text{coefficient of } x^{n-i}}{\text{leading coefficient}} = (-1)^i \dfrac{a_i}{a_0}$.

Ex: $x^3 - 3x + 7 = 0$: $S_1 = 0,\ S_2 = -3,\ S_3 = -7$.

Grade - Medium

Practice map, main concept: Sheet: J-I 28; J-II 13, 16; J-M 11; C-I 13 | DPP: 5(2, 8) | Question Bank: 19

Also used in: Sheet: J-II 14-15; J-A 8

Question 60

How do you find the sum of squares of the roots?

Model Answer

$$\sum \alpha_i^2 = S_1^2 - 2S_2$$

Ex: $x^5 - 7x^4 + 3x^3 + 4x - 5 = 0$: $S_1 = 7,\ S_2 = 3 \Rightarrow \sum x_i^2 = 49 - 6 = 43$.

If $\alpha + \beta + \gamma = 0$ then $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$. Ex: $x^3 - 3x + 7 = 0$: $\alpha^3 + \beta^3 + \gamma^3 = 3(-7) = -21$.

Grade - Medium

Practice map, main concept: Sheet: J-A 8; C-I 16; C-II 14

Also used in: Sheet: C-I 13

Location of Roots

Question 61

Are you able to write the conditions when conditions on roots are specified?

Model Answer

Use three tools: D, the vertex $x = -\frac{b}{2a}$ against the bounds, and the sign of $a\,f(p)$ at the given points.

Both roots $> d$: $D \geq 0,\ af(d) > 0,\ -\frac{b}{2a} > d$. Both $< d$: same with $-\frac{b}{2a} < d$. Both in (d, e): $D \geq 0,\ af(d) > 0,\ af(e) > 0,\ d < -\frac{b}{2a} < e$.

d lies between the roots: $af(d) < 0$. One root $> e$ and the other $< d$ $(d < e)$: $af(d) < 0$ and $af(e) < 0$.

Ex: both roots of $x^2 - 6kx + (2 - 2k + 9k^2) = 0$ exceed 3: $D \geq 0 \Rightarrow k \geq 1$; $3k > 3 \Rightarrow k > 1$; $f(3) > 0 \Rightarrow k < 1$ or $k > \frac{11}{9}$. Hence $k > \dfrac{11}{9}$.

Ex: roots of $x^2-(k+1)x+k^2+k-8 = 0$ on either side of 2: $f(2) < 0 \Rightarrow k^2-k-6 < 0 \Rightarrow k \in (-2, 3)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 30-33; J-II 17-20, 23-24, 35, 42, 46; J-M 6; C-I 3-4, 8; C-II 6 | DPP: 6(2-8) | Question Bank: 1, 17

Also used in: Sheet: J-II 22, 25

Question 62

When will we not apply conditions on D and $-\dfrac{b}{2a}$?

Model Answer

When a given point lies between the roots ($af(d) < 0$ already forces $D > 0$, so D and the vertex need no check). This covers one root on each side of a number and the case of one root $> e$, other $< d$.

If D is a perfect square, compute the roots directly and apply the condition.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 31, 33 | DPP: 6(2, 5)

Question 63

When exactly one root lies in an interval (d, e), what is the condition, and how do you handle the endpoint cases?

Model Answer

$f(d)\,f(e) < 0$. Then check separately the cases $f(d) = 0$ and $f(e) = 0$: find the other root and see whether it lies in the interval.

Ex: $x^2 - (k + 1)x + 2k = 0$, exactly one root in $(0, 3)$: $f(0)f(3) = 2k(6 - k) < 0 \Rightarrow k < 0$ or $k > 6$. $k = 0$: roots $0, 1$ (one root in $(0, 3)$) is valid. $k = 6$: roots $3, 4$ (none) is rejected. So $k \in (-\infty, 0] \cup (6, \infty)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 64

How do you use the Intermediate Value Theorem ($f(p)f(q) < 0$) in questions on quadratic equations?

Model Answer

f is continuous, so if $f(p)f(q) < 0$ then $f(x) = 0$ has a root in (p, q); for a quadratic exactly one. This proves real roots without computing D.

Ex: if $(a + b + c)(4a + 2b + c) < 0$ then $f(1)f(2) < 0$ for $f(x) = ax^2 + bx + c$, so the equation has real roots.

Ex: $y = \dfrac{(x-a)(x-c)}{(x-b)(x-d)}$ with $a > b > c > d$: $(x-a)(x-c) - y(x-b)(x-d) = 0$ has $f(b) = (b-a)(b-c) < 0$ and $f(d) = (d-a)(d-c) > 0$ for every y, so a real x exists for every y and the range is $\mathbb{R}$.

Grade - Low

Practice map, main concept: Sheet: C-II 10

Also used in: Sheet: J-I 33

Question 65

How do you find the number of real solutions of an equation like $x^4 + px^2 + q = 0$ (a quadratic in x^2)?

Model Answer

Put $t = x^2$. A negative root t gives no real x, $t = 0$ gives one, and each positive t gives two. Apply location-of-roots conditions on the quadratic in t relative to 0.

Four real solutions: both roots in t positive and distinct: $D > 0,\ -\frac{b}{2a} > 0,\ af(0) > 0$. Two: roots of opposite sign, or both positive and equal. Three: $t = 0$ and one positive root. One: $t = 0$ is a double root, or $t = 0$ with a negative root. None: imaginary t or both negative.

Ex: $x^4 + (1-2k)x^2 + k^2 - 1 = 0$ has four real solutions: $D > 0 \Rightarrow k < \frac{5}{4}$; sum of t-roots $2k - 1 > 0$; product $k^2 - 1 > 0 \Rightarrow k > 1$ or $k < -1$. So $k \in \left(1, \dfrac{5}{4}\right)$. For $k = 1$: $t = 0, 1$, which gives three solutions $0, \pm 1$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 66

What are the conditions for a quadratic (with $a > 0$) to have at least one positive root, or to be positive for all $x > 0$?

Model Answer

At least one positive root ($a > 0$): $f(0) < 0$, or ($D \geq 0,\ -\frac{b}{2a} > 0,\ f(0) \geq 0$). Similarly for at least one negative root with $-\frac{b}{2a} < 0$.

$f(x) > 0$ for all $x > 0$ ($a > 0$): either $D < 0$, or ($D \geq 0$, $-\frac{b}{2a} \leq 0$ and $f(0) \geq 0$ with no positive root).

Ex: $x^2 + kx + 4 > 0$ for all $x > 0$: $-4 < k < 4$ from $D < 0$, and $k \geq 4$ from the second case ($-\frac{k}{2} \leq 0$, $f(0) = 4 > 0$). Hence $k > -4$ ($k = -4$ fails at $x = 2$).

Grade - Low

Practice map, main concept: Sheet: J-II 21

Also used in: none

Miscellaneous

Question 67

How do you solve a problem on an infinite continued radical or fraction, such as $x = \sqrt{a + \sqrt{a + \cdots}}$ or $x = a + \dfrac{1}{a + \frac{1}{a+\cdots}}$?

Model Answer

The pattern repeats, so the expression equals x inside itself: $x = \sqrt{a + x}$ or $x = a + \dfrac{1}{x}$. Form the quadratic and reject the root that violates the sign or positivity condition.

Ex: $x = \sqrt{6 + \sqrt{6 + \cdots}}$: $x^2 = x + 6 \Rightarrow x = 3$ or -2; $x > 0$, so $x = 3$.

Grade - Medium

Practice map, main concept: DPP: 5(6-7) | Question Bank: 12

Also used in: none

Question 68

Write the factors of $a^3 + b^3 + c^3 - 3abc$ and use $a^3 + b^3 + c^3 = 3abc$.

Model Answer

$$a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = \tfrac{1}{2}(a+b+c)\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

So $a^3 + b^3 + c^3 = 3abc \iff a + b + c = 0$ or $a = b = c$.

Ex: $(x-2)^3+(x-3)^3+(5-2x)^3 = 0$: the bases add to 0, so $3(x-2)(x-3)(5-2x) = 0 \Rightarrow x = 2,\ 3,\ \dfrac{5}{2}$.

Grade - Low

Practice map, main concept: Sheet: J-II 14

Also used in: none

Question 69

What do you keep in mind if you have squared while solving an equation?

Model Answer

Squaring can introduce extraneous roots. Substitute every root back, and check the domain and the sign of any square root (e.g. $\sqrt{\ } \geq 0$).

Ex: $x = \sqrt{x + 2}$: squaring gives $x^2 - x - 2 = 0 \Rightarrow x = 2, -1$; $x = -1$ is rejected because $\sqrt{\ } \geq 0$. So $x = 2$.

Grade - Medium

Practice map, main concept: Sheet: J-II 29; J-M 3 | DPP: 5(3) | Question Bank: 15

Also used in: Sheet: J-II 28; C-II 6

Question 70

How do you solve the equation $ax^4 + bx^3 + cx^2 \pm bx + a = 0$?

Model Answer

Divide by x^2. For $+bx$: put $t = x + \dfrac{1}{x}$, giving $at^2 + bt + (c - 2a) = 0$. For $-bx$: put $t = x - \dfrac{1}{x}$, giving $at^2 + bt + (c + 2a) = 0$.

Ex: $x^4 + x^3 - 4x^2 + x + 1 = 0$: $t^2 + t - 6 = 0 \Rightarrow t = 2$ or -3. $t = 2 \Rightarrow x = 1$ (double); $t = -3 \Rightarrow x^2 + 3x + 1 = 0 \Rightarrow x = \dfrac{-3 \pm \sqrt{5}}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 25; J-M 4 | Question Bank: 16

Also used in: Sheet: C-I 17

Question 71

Do you know how to apply Descartes' rule of signs?

Model Answer

The number of positive roots is at most the number of sign changes in the coefficients of $f(x)$, and differs from it by an even number. For negative roots apply the rule to $f(-x)$.

Ex: $x^7 + 5x^3 + 3x + 1$: no sign change, so no positive root. $f(-x) = -x^7 - 5x^3 - 3x + 1$ has one sign change, so exactly one negative root. Hence the equation has 1 real solution.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 72

How do you solve an equation that reduces to a quadratic by a substitution such as $t = x + \frac{1}{x}$, $t = x^2$, $t = e^x$, $t = a^x$, and how do you restrict the range of t?

Model Answer

Solve the quadratic in t, keep only t in its allowed range, and count the x for each kept t. Ranges: $t = x + \frac{1}{x} \Rightarrow |t| \geq 2$ (x real); $t = x^2 \Rightarrow t \geq 0$; $t = e^x$ or a^x $(a > 0) \Rightarrow t > 0$.

Ex: $\left(x^2 + \dfrac{1}{x^2}\right) + 3\left(x + \dfrac{1}{x}\right) + 5 = 0$: $t = x + \frac{1}{x}$ gives $t^2 + 3t + 3 = 0$ with $D < 0$, so no real solution.

Ex: $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0$: divide by e^{2x}, put $z = e^x - e^{-x}$: $z^2 + 8z + 15 = 0 \Rightarrow z = -3, -5$. Each gives one positive $t = e^x$ (less than 1), so 2 real solutions, both negative.

Ex: $(\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x = 10$: the bases have product 1, so $t + \frac{1}{t} = 10 \Rightarrow t = 5 \pm 2\sqrt{6} = (\sqrt{3} \pm \sqrt{2})^2 \Rightarrow x = \pm 2$.

Grade - High

Practice map, main concept: Sheet: J-II 28; J-M 1, 7, 17; C-I 5, 17, 21 | Question Bank: 13, 21

Also used in: Sheet: J-II 25, 29-30; J-M 4; C-I 24; C-II 4 | DPP: 3(9-10); 4(2); 5(3) | Question Bank: 16

Question 73

How do you solve an equation that is homogeneous in two expressions u and v, such as $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$?

Model Answer

Divide by v^2 (check $v = 0$ separately), put $t = \dfrac{u}{v}$, solve the quadratic in t, then solve $u = tv$ for each t.

Ex: $u = x^2 + 2$, $v = x$: $t^2 - 6t + 8 = 0 \Rightarrow t = 2, 4$. $t = 2$: $x^2 - 2x + 2 = 0 \Rightarrow x = 1 \pm i$. $t = 4$: $x^2 - 4x + 2 = 0 \Rightarrow x = 2 \pm \sqrt{2}$.

Grade - Low

Practice map, main concept: DPP: 4(2)

Also used in: none

Question 74

How do you solve an equation that is a product of four linear factors, such as $(x+9)(x-3)(x-7)(x+5) = 385$?

Model Answer

Pair the factors so that the constants in each pair have the same sum (then the two products have the same x-terms). Put the common quadratic expression $= t$, solve the quadratic in t, then solve for x.

Ex: $(x+9)(x-7) = x^2+2x-63$ and $(x-3)(x+5) = x^2+2x-15$. With $t = x^2+2x$: $(t-63)(t-15) = 385 \Rightarrow t^2-78t+560 = 0 \Rightarrow t = 8$ or 70. So $x = -4,\ 2$ or $x = -1 \pm \sqrt{71}$.

Grade - Low

Practice map, main concept: Sheet: J-II 30

Also used in: none

Question 75

How do you find the roots of a composite equation $p(p(x)) = 0$?

Model Answer

$p(p(x)) = 0 \iff p(x) = r$ where r is a root of p. Solve $p(x) = r_1$ and $p(x) = r_2$ separately and examine the nature of each solution.

Ex: $p(x) = x^2+1$ has purely imaginary roots $\pm i$. Then $p(x) = \pm i \Rightarrow x^2 = -1 \pm i$: x is not real (as x^2 is not a real number) and not purely imaginary (the square of a purely imaginary number is real). So $p(p(x)) = 0$ has neither real nor purely imaginary roots.

Grade - Medium

Practice map, main concept: Sheet: J-A 2

Also used in: none